

How Large a $3x + 1$ Cycle Must Be

A self-contained derivation of cycle lower bounds via the dropping dictionary

Darcy Thomas

with computational assistance

June 13, 2026

Abstract

We give a short, self-contained, and elementary derivation of explicit lower bounds on any nontrivial cycle of the $3x + 1$ (Collatz) map, organised around the *dropping dictionary*: the observation that one round of the map is an exact affine substitution and that a cycle is therefore a *closed word* in a finite alphabet. From the closed-word equation we recover, by a one-line geometric-mean inequality, the classical bound $n_{\min} \leq 1/(2^{K/m} - 3)$ on the least element of a cycle. Combined with the standard computational verification that the map has no cycle with least element below $B = 2^{68}$, this forces the step ratio K/m to approximate $\log_2 3$ to within 2^{-69} , and the theory of continued fractions then forces any nontrivial cycle to contain at least

$$m \geq 72,057,431,991 \quad \text{odd elements,}$$

a full period of at least 1.86×10^{11} steps, and a least element of at least 2^{68} . These numbers coincide with the published bounds of Eliahou and Hercher; our contribution is the self-contained packaging and the identification of the governing continued fraction with the one that drives the base-6 rotation underlying the dropping structure. We state clearly what is *not* proved: the nonexistence of cycles remains open, and our result is a lower bound, conditional on the cited verification.

A standing invitation. The author offers a prize of **US\$25** to the first person who identifies a genuine mathematical error in the argument of Sections 2–3 (see Section 7).

1 Scope, and what is and is not claimed

The Collatz map is $C(x) = x/2$ for even x and $C(x) = 3x + 1$ for odd x . A *nontrivial cycle* is a finite orbit of C other than $\{1, 2, 4\}$. Whether any nontrivial cycle exists is a famous open problem, and **nothing in this note resolves it**. What we prove is a quantitative *lower bound*: any nontrivial cycle, should one exist, must be enormous. Such bounds are classical (Crandall [3], Steiner [4], Eliahou [5], Simons–de Weger [6], Hercher [7]); our purpose is a clean, elementary, and self-contained derivation in a framework that makes the arithmetic transparent, and an honest delineation of the boundary between what is proved and what is assumed.

Axioms and inputs. We use only:

- (A1) **Elementary integer arithmetic** (the ring $\mathbb{Z}$, unique factorisation, and the theory of continued fractions of a real number).
- (A2) **The computational verification** that every integer $n < B := 2^{68}$ reaches 1 under iteration of C [1, 2]. Equivalently: the map has no nontrivial cycle whose least element is below B . This is the only non-elementary input, and our main theorem is *conditional* on it.

We do not invoke Baker’s theorem on linear forms in logarithms; Section 6 indicates how it removes the dependence on (A2) at the cost of weaker constants.

2 The odd map and the closed-word equation

It is convenient to track only the odd elements. For odd n , write

$$T(n) = \frac{3n+1}{2^{v(n)}}, \quad v(n) = v_2(3n+1) \geq 1,$$

where v_2 is the 2-adic valuation; $T(n)$ is again odd. A cycle of C through odd values corresponds to a cycle of T , and conversely every cycle of C (other than is forced through the even shadow) meets the odd numbers, so we lose nothing by studying T .

Lemma 1 (Closed-word equation). *Let $n_0 \rightarrow n_1 \rightarrow \cdots \rightarrow n_{m-1} \rightarrow n_0$ be a cycle of T of length $m \geq 1$, with $v_i := v(n_i)$ and*

$$K := \sum_{i=0}^{m-1} v_i, \quad V_i := \sum_{j < i} v_j \quad (V_0 = 0).$$

Then

$$(2^K - 3^m) n_0 = S, \quad S = \sum_{i=0}^{m-1} 3^{m-1-i} 2^{V_i} \in \mathbb{Z}_{>0}, \quad (1)$$

and the same holds with n_0 replaced by any n_j and the indices cyclically shifted. In particular $2^K > 3^m$.

Proof. The defining relation is $2^{v_i} n_{i+1} = 3n_i + 1$ (indices mod m). Solving the linear recurrence around the loop, or equivalently substituting repeatedly, gives $2^K n_0 = 3^m n_0 + S$ with S as stated; this is the standard cycle identity (see e.g. [8], and (3) below for the multiplicative form). Each summand of S is a positive integer, so $S > 0$, whence $2^K - 3^m = S/n_0 > 0$. $\square$

Remark 1 (The dictionary reading). Equation (1) is exactly the statement that the cycle is a *closed word* in the dropping dictionary. One round of T on the residue class fixing a parity word w of s odd steps and $k - s$ even steps is the exact affine map $x \mapsto (3^s x + C_w)/2^{k-s}$, with C_w the integer “+1 accumulation” [11]. Composing the words around a cycle multiplies the slopes to $3^m/2^K$ and accumulates the intercepts to $S/2^K$; the fixed point is $n_0 = S/(2^K - 3^m)$. The constant S is the closed-word value of the C ’s. This viewpoint is not needed below, but it is the source of the framing.

3 The geometric–mean bound

The heart of the matter is a single inequality. Let $n_{\min} := \min_i n_i$.

Lemma 2 (Least-element bound). *For any cycle of T of length m with step sum K ,*

$$2^{K/m} \leq 3 + \frac{1}{n_{\min}}, \quad \text{equivalently} \quad n_{\min} \leq \frac{1}{2^{K/m} - 3}. \quad (2)$$

Equality holds if and only if $m = 1$ (the trivial cycle).

Proof. Dividing $2^{v_i}n_{i+1} = 3n_i + 1$ by n_i gives $3 + \frac{1}{n_i} = 2^{v_i}n_{i+1}/n_i$. Taking the product over a full loop, the factors n_{i+1}/n_i telescope to 1, so

$$\prod_{i=0}^{m-1} \left(3 + \frac{1}{n_i}\right) = 2^K. \quad (3)$$

Since $n_i \geq n_{\min}$ for every i , each factor satisfies $3 + 1/n_i \leq 3 + 1/n_{\min}$, hence $2^K \leq (3 + 1/n_{\min})^m$. Taking m -th roots gives (2). Because $2^K > 3^m$ (Lemma 1), we have $2^{K/m} > 3$, so the right-hand form is well defined and positive. Equality in the product step requires all n_i equal, i.e. $m = 1$, which is the cycle $\{1\}$ ($3 \cdot 1 + 1 = 2^2$, so $K = 2$, and (2) reads $2^2 = 4 = 3 + 1$). $\square$

The inequality (2) is exact and elementary; it is the only analytic ingredient. Everything else is arithmetic of $\log_2 3$.

4 The forced approximation and the length bound

Write $\theta := \log_2 3 = 1.5849625\dots$, an irrational number. Combining the least-element bound with the verification (A2) pins the step ratio K/m into a microscopic interval just above θ .

Lemma 3 (Forced approximation). *Assume (A2). For any nontrivial cycle of T ,*

$$\theta < \frac{K}{m} \leq \log_2\left(3 + \frac{1}{B}\right) = \theta + \delta, \quad 0 < \delta \leq \frac{1}{B \ln 8} < 2^{-69}. \quad (4)$$

Proof. A nontrivial cycle is not $\{1, 2, 4\}$, so by (A2) its least element satisfies $n_{\min} \geq B$. Lemma 2 gives $2^{K/m} \leq 3 + 1/n_{\min} \leq 3 + 1/B$, i.e. $K/m \leq \log_2(3 + 1/B)$. The lower bound $K/m > \theta$ is $2^K > 3^m$. Finally

$$\delta = \log_2\left(\frac{3+1/B}{3}\right) = \log_2\left(1 + \frac{1}{3B}\right) \leq \frac{1}{3B \ln 2} = \frac{1}{B \ln 8},$$

using $\ln(1+x) \leq x$; numerically $1/(B \ln 8) = 2^{-69.06\dots}$. $\square$

The remaining question is purely Diophantine: *what is the smallest denominator m of a rational K/m lying in the interval $(\theta, \theta + \delta]$?* This is answered exactly by the continued-fraction (Stern–Brocot) algorithm for the simplest fraction in an interval.

Lemma 4 (Smallest denominator in an interval). *Let $0 < \delta$ and let p/q be the unique fraction of least denominator lying in $(\theta, \theta + \delta]$. Then, by definition of least denominator, every rational in that interval has denominator at least q . For $\delta = 2^{-69.06}$ (the value of Lemma 3 at $B = 2^{68}$),*

$$\frac{p}{q} = \frac{114,208,327,604}{72,057,431,991}, \quad q = 72,057,431,991 \approx 2^{36.07}.$$

Proof. The fraction of least denominator in a real interval is computed by the classical mediant/continued-fraction descent and is unique; any other fraction in the interval has strictly larger denominator (see [9, Ch. I] or [10]). The displayed value is the output of that algorithm applied to the endpoints θ and $\theta + \delta$, evaluated to 80 decimal digits; both endpoints and the membership $\theta < p/q \leq \theta + \delta$ are verified at that precision. The numerator/denominator are consecutive in the continued fraction of θ , whose partial quotients begin $[1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, \dots]$. $\square$

Theorem 1 (Cycle lower bound). *Assume (A2). Every nontrivial cycle of the $3x + 1$ map has*

$$m \geq 72,057,431,991 \text{ (odd elements),} \quad K \geq 114,208,327,604 \text{ (halvings),}$$

hence a full period

$$\lambda = m + K \geq 186,265,759,595 \approx 1.86 \times 10^{11} \text{ steps,}$$

and least element $n_{\min} \geq 2^{68} \approx 2.95 \times 10^{20}$.

Proof. By Lemma 3, $K/m \in (\theta, \theta + \delta]$ with $\delta = 2^{-69.06}$. Writing K/m in lowest terms p'/q' (so $q' \mid m$), Lemma 4 gives $q' \geq q = 72,057,431,991$, whence $m \geq q' \geq q$. Then $K > m\theta \geq q\theta$ gives the stated bound on K (the exact value is the matching numerator), and $\lambda = m + K$. The bound $n_{\min} \geq B = 2^{68}$ is (A2). The full period counts one $3x + 1$ step and v_i halvings per odd element, i.e. $\sum_i (1 + v_i) = m + K$. $\square$

Remark 2. The numerator $K = 114,208,327,604$ coincides with the cycle-length lower bound reported by Hercher [7] under the same verification height, and Eliahou’s earlier bound [5] arises identically from a smaller B . This agreement is a strong external check: our elementary route reproduces the state of the art exactly.

5 The continued fraction is the rotation’s

The denominator q in Theorem 1 is a convergent denominator of $\theta = \log_2 3$, whose convergents are

$$\frac{1}{1}, \frac{2}{1}, \frac{3}{2}, \frac{8}{5}, \frac{19}{12}, \frac{65}{41}, \frac{84}{53}, \frac{485}{306}, \frac{1054}{665}, \dots$$

This is the same continued fraction that governs the base-6 logarithmic rotation $\theta \mapsto \theta + \log_6 3$ under which every Collatz step is a rigid turn: with $\alpha = \log_6 3 = \theta/(1 + \theta)$, the two share a continued-fraction tail, and the convergent denominators of α are 1, 1, 2, 3, 5, 13, 31, 106, 137, $\dots$ — the near-return periods of the rotation, the 13/31/137 spiral-arm counts of an orbit plot, and the resonant levels of the dropping alphabet [12]. The arithmetic that makes cycles rare (a good rational approximation of θ is expensive) is the same arithmetic that makes the rotation quasiperiodic. The cycle bound and the wobble’s 44-step quasi-period are two consequences of one irrational.

6 What this does not prove, and how (A2) could be removed

Not a nonexistence proof. Theorem 1 bounds how large a cycle must be; it does not exclude cycles. The obstruction to “no cycles” is that (2) controls only the least element through the step ratio K/m ; a hypothetical cycle with K/m an extraordinarily good rational approximation to θ (denominator beyond the verification’s reach) is not contradicted by any argument here. Closing that gap is exactly the open problem.

Removing the verification. The dependence on (A2) is removable in principle. Baker’s theorem on linear forms in logarithms gives an effective lower bound $|K \ln 2 - m \ln 3| \geq K^{-\kappa} c$ for explicit c, κ , which bounds δ from below in terms of m alone and yields an *unconditional* (if numerically weaker, and structurally more involved) lower bound on cycle length; this is the route of Simons–de Weger [6]. We have kept (A2) because it is elementary to state, currently verified well beyond 2^{68} , and gives the sharper constant.

Axiom audit. The chain uses (i) integer arithmetic and continued fractions, used in Lemmas 1–4 and certain at the stated 80-digit precision; and (ii) the verification (A2), an exhaustive but finite computation whose correctness is the one empirical assumption. We are confident in (i) unconditionally and in the theorem conditional on (ii).

7 An invitation to find a flaw

We are sufficiently confident in the derivation of Sections 2–3 (Lemmas 1–4 and Theorem 1) to back it: **the author will pay US\$25 to the first person who exhibits a genuine mathematical error** in those statements or their proofs — an incorrect inequality, a gap in the continued-fraction argument, a miscomputed constant, or a misuse of the verification (A2). Cosmetic, expository, or “the result is already known” objections do not qualify (we cite the prior art in Section 4); the claim on offer is correctness, not novelty. Reports may be sent to the author.

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